

DBMS — Lecture 1

Entity-Relationship (ER) Model

Learning Objectives

Understand the purpose of the ER model.

Identify entities, attributes, relationships, and keys.

Understand basic ER diagram notation.

ER Model

The **Entity-Relationship (ER) model** is a conceptual data model used to represent real-world objects and the relationships among them before implementing a database.

It is widely used during database design because it gives a clear, visual representation of requirements.

Core Concepts

Entity: A distinguishable real-world object, such as Student or Course.

Entity Set: Collection of similar entities.

Attribute: Property of an entity, such as Student_ID or Name.

Relationship: Association between entities, such as Student *enrolls in* Course.

Key Attribute: Attribute that uniquely identifies an entity.

Common ER Notation

Rectangle: Entity set.

Oval: Attribute.

Diamond: Relationship.

Underlined attribute: Key attribute.

Double oval: Multivalued attribute.

Double rectangle: Weak entity.

Example

STUDENT — *enrolls* — **COURSE**

Student attributes: Student_ID, Name, Branch. Course attributes: Course_ID, Course_Name.

Key Takeaway

ER modeling converts real-world requirements into a conceptual database design.